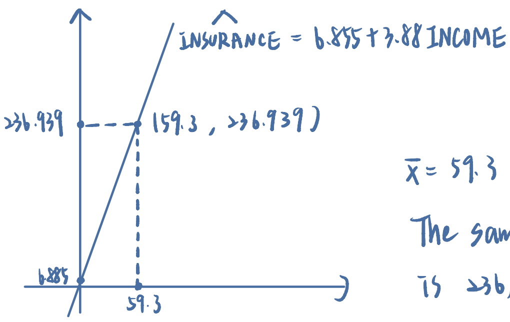
3.18 A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let *INCOME* be household income (thousands of dollars) and *INSURANCE* the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

$$\widehat{INSURANCE} = \underset{(7.383)}{6.855} + \underset{(0.112)}{3.880}INCOME \qquad \text{(se)}$$

a. Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of $INCOME = 59.3$. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.



$\bar{x} = 59.3 \Rightarrow \bar{y} = 6.855 + 3.88 \times 59.3 = 236.939$

The sample mean of the amount of life insurance held is 236,939 dollars.

b. How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.

$df = 20 - 2 = 18$, $\alpha = 0.025$, $t_{(0.025,\,18)} = 2.101$

The 95% C.I $= 3.88 \pm t_{(0.025,\,18)}\, se(b_2) = 3.88 \pm 2.101 \times 0.112 = (3.6447,\ 4.1153)$

In repeating sampling, about 95% intervals constructed this way will cover the true value of β_2.

c. Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is -0.746.

$Var(\hat{y}) = Var(b_1) + x^2 var(b_2) + 2x\, cov(b_1, b_2) = (7.383)^2 + 100^2 \times (0.112)^2 + 2 \times 100 \times (-0.746) = 30.748689$

$se(\hat{y}) = \sqrt{30.748689} = 5.54515$. $\hat{y} = 6.855 + 3.88 \times 100 = 394.855$, $\alpha = 0.01 \Rightarrow t_{(0.005,\,18)} = 2.878$

99% CI $= 394.855 \pm t_{(0.005,18)}\, se(\hat{y}) = 394.855 \pm 2.878 \times 5.54515 = (378.8961,\ 410.8139)$ #

d. One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.

$H_0: \beta_2 = 5 \quad H_1: \beta_2 \neq 5$

$\alpha = 0.05$

$t = \dfrac{\hat{\beta}_2 - 5}{se(\hat{\beta}_2)} \sim t_{(18)} \Rightarrow t = \dfrac{3.88 - 5}{0.112} = -10$

reject region $= t_{(\frac{\alpha}{2},\, df)} = t_{(0.025,\,18)} = 2.101$

$t = |-10| > 2.101 \Rightarrow$ reject H_0.

The data provides sufficient evidence to reject the claim, as the estimated marginal increase of \$3880 is significantly less than hypothesised \$5000 at the 95% confidence level.

e. Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

$H_0: \beta_2 = 1 \quad H_1: \beta_2 > 1$

$\alpha = 0.01$

$t = \frac{\hat{\beta}_2 - 1}{0.112} = \frac{3.88-1}{0.112} = 25.714$

reject region $= t_{(0.01, 18)} = 2.552$

$t = 25.714 > 2.552 \Rightarrow$ reject H_0.

We reject the null hypothesis and conclude that the amount of life insurance held increases by significantly more than the increase in household income at the 99 % confidence level. #

3.23 The data file *collegetown* contains data on 500 single-family houses sold in Baton Rouge, Louisiana, during 2009–2013. The data include sale price in \$1000 units, *PRICE*, and total interior area in hundreds of square feet, *SQFT*.

a. Using the quadratic regression model, $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$, test the hypothesis that the marginal effect on expected house price of increasing the size of a 2000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test *p*-value. What do you conclude?

marginal effect : $\frac{\partial PRICE}{\partial SQFT} = 2\alpha_2 \cdot SQFT$

$H_0: 2 \times 20 \times \alpha_2 \le 13$, $H_1: 2 \times 20 \times \alpha_2 > 13$

$\alpha = 0.05$, $t_0 = \frac{40\hat{\alpha}_2 - 13}{se(\hat{\alpha}_2)} \sim t_{(500-2)}$

(with $\frac{13}{40}$ written above the 13)

reject region $= t_0 > t_{(0.05, 498)} = 1.648$

$t_0 = -26.72875$

p-value $= P(t > -26.72875 \mid H_0 \text{ true}) = 1 > \alpha = 0.05$

$\Rightarrow$ Don't reject H_0.

The calculated test statistic is t_0 = −26.73, which is significantly smaller than the critical value t_c = 1.648. Consequently, the p−value is 1.
Conclusion: Do not reject H_0. There is no statistical evidence to suggest that the marginal effect of an additional 100 sqft for a 2,000 sqft house exceeds \$13,000. The data suggests that for smaller homes, the value added by extra space is relatively low.

b. Using the quadratic regression model in part (a), test the hypothesis that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000.

$H_0: 2 \times 40 \times \alpha_2 \le 13$, $H_1: 2 \times 40 \times \alpha_2 > 13$

$t_0 = 4.189467$

reject region $= t_0 > t_{(0.05, 498)} = 1.648$

p-value $= P(t > 4.189467 \mid H_0 \text{ is true}) = 1.65395 \times 10^{-5} < \alpha = 0.05$, reject H_0.

For a 4,000 sqft house, the test statistic is t_0 = 4.19, which exceeds the critical value of 1.648. The p−value is approximately 0.0000165 which is much smaller than the significance level alpha = 0.05
Conclusion: Reject H_0. We conclude that the marginal effect for a 4,000 sqft house is significantly greater than \$13,000. This reflects the quadratic nature of the model, where the value of additional space increases as the total house size grows.

c. Using the quadratic regression model in part (a), estimate the expected price $E(PRICE|SQFT) = \alpha_1 + \alpha_2 SQFT^2$ for a house of 2000 square feet. Construct a 95% interval estimate of the expected price. Describe your interval estimate to a general audience.

E(Price | SQFT = 20) = 167.3735

95% C.I = (158.0481 , 176.6988)

The predicted expected price for a house with 2,000 sqft (SQFT = 20) is $167,373.50.
Confidence Interval: The 95% confidence interval for the mean price is [158048.1, 176698.8].
Conclusion: In repeated sampling, about 95% intervals constructed this way will cover the true value of the expected price.

d. Locate houses in the sample with 2000 square feet of living area. Calculate the sample mean (average) of their selling prices. Is the sample average of the selling price for houses with $SQFT = 20$ compatible with the result in part (c)? Explain.

Sample mean of their selling price is 163.333, withing (158.0481, 176.6988)

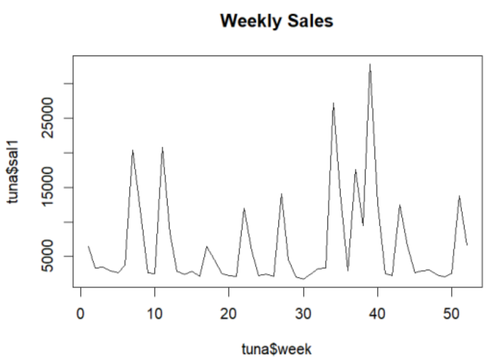
This is compatible with the result in part (c).

The actual sample mean price for houses in the dataset that are exactly 2,000 sqft is $163,333.
Conclusion: Since the sample mean of $163,333.30 falls within the 95% confidence interval obtained in part (c), the observed data is compatible with the model's prediction. This indicates that the quadratic model provides a reliable fit for houses of this size.

3.31 Data on weekly sales of a major brand of canned tuna by a supermarket chain in a large midwestern U.S. city during a mid-1990s calendar year are contained in the data file tuna. There are 52 observations for each of the variables. The variable SAL1 = unit sales of brand no. 1 canned tuna, and APR1 = price per can of brand no. 1 tuna (in dollars).

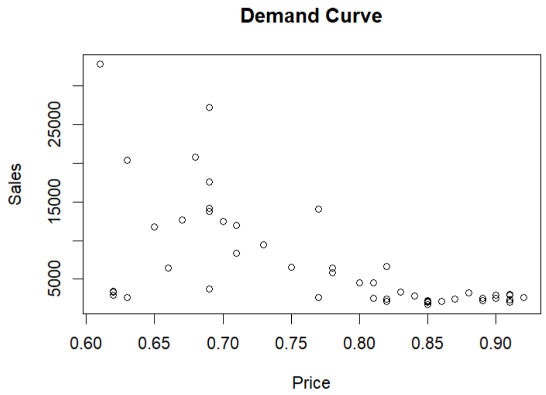
a. Calculate the summary statistics for *SAL*1 and *APR*1. What are the sample means, minimum andmaximum values, and their standard deviations. Plot each of these variables versus *WEEK*. How much variation in sales and price is there from week to week?

```
> # (a)
> summary(tuna$sal1)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   1762    2485    3136    6719    8612   32820
> sd(tuna$sal1)
[1] 6915.083
>
> summary(tuna$apr1)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.6100  0.6900  0.8100  0.7825  0.8625  0.9200
> sd(tuna$apr1)
[1] 0.09803711
```



The weekly sales (sal1) with a mean of 6,719 units and a high standard deviation of 6,915.08, ranging from a minimum of 1,762 to a maximum of 32,820 units. In contrast, the price (apr1) remains relatively stable, with a mean of $0.78 and a standard deviation of $0.098, fluctuating between $0.61 and $0.92. The plots reveal significant weekly fluctuations; the sharp spikes in weekly sales consistently coincide with the sharp troughs in weekly price. This indicates that the supermarket frequently employs discounting strategies, which in turn trigger dramatic ~~surges~~ increases in consumer demand.

b. Plot the variable SAL1 (y-axis) against APR1 (x-axis). Is there a positive or inverse relationship? Is that what you expected, or not? Why?



The scatter plot of SAL1 (Sales) against APR1 (Price) reveals a negative relationship. As the price per can increases from approximately \$0.60 to \$0.92, the weekly unit sales tend to decrease and stabilize at a lower level. Conversely, the highest sales volumes are exclusively observed at lower price points. This pattern is precisely what is expected according to the **Law of Demand**, which posits that, all else being equal, the quantity demanded of a good falls when the price of the good rises. In this specific case, the high density of points at higher prices with low sales suggests that consumers are **highly sensitive** to price increases for this brand of canned tuna.

c. Create the variable PRICE1= 100APR1. Estimate the linear regression $SAL1 = \beta_1 + \beta_2 PRICE1 + e$. What is the point estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1? What is a 95% interval estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1?

```
> summary(model_c)

Call:
lm(formula = sal1 ~ price1, data = tuna)

Residuals:
     Min       1Q   Median       3Q      Max 
-10869.5  -1588.3   -499.3   1626.2  18607.1 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 40714.22    6196.42   6.571 2.82e-08 ***
price1       -434.45      78.58  -5.528 1.17e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5502 on 50 degrees of freedom
Multiple R-squared:  0.3794,	Adjusted R-squared:  0.367 
F-statistic: 30.56 on 1 and 50 DF,  p-value: 1.172e-06
```

The point estimate for the effect of a one-cent increase in price is -434.45, indicating that for every one-cent increase in the price of brand no. 1 tuna, the expected weekly sales decrease by approximately 434 units. This effect is statistically significant at the 1% level, as evidenced by the very small p-value (1.17×10^{-6}).

```
> confint(model_c, "price1", level=0.95) # 95% 置信區間
           2.5 %    97.5 %
price1 -592.2897 -276.6049
```

The 95% interval estimate for this effect ranges from -592.29 to -276.60. This means we are 95% confident that the true reduction in weekly sales resulting from a one-cent price hike lies between approximately 277 and 592 units.

d. Construct a 90% interval estimate for the expected number of cans sold in a week when the price per can is 70 cents.

```
       fit      lwr      upr
1  10302.9 8624.934 11980.87
```

When the price per can is set to 70 cents, the regression model predicts an expected weekly sales volume of approximately 10,303 units (the fit value). Furthermore, the 90% interval estimate for the expected number of cans sold ranges from 8,625 to 11,981 units. In the context of repeated sampling, about 90% of all such constructed intervals would be expected to contain the true population parameter for the expected weekly sales at this price point.

e. Construct a 95% interval estimate of the elasticity of sales of brand no. 1 with respect to the price of brand no. 1 "at the means." Treat the sample means as constants and not random variables. Do you find the sales are fairly elastic, or fairly inelastic, with respect to price? Does this make economic sense? Why?

```
> print(elasticity) # elasticity
   price1
-5.059825
> print(ci_elasticity) #  95% Confidence Interval
            2.5 %    97.5 %
price1 -6.898149 -3.221501
```

The point estimate for the price elasticity of demand at the sample means is approximately -5.06, with a 95% confidence interval of [-6.90, -3.22]. Since the absolute value of the elasticity is substantially **greater than 1**, we conclude that the consumer demand for this brand of tuna is **highly elastic**. This indicates that consumers are extremely sensitive to price changes; specifically, a 1% increase in price is expected to lead to a 5.06% decrease in the quantity demanded. This high degree of elasticity makes economic sense given the availability of numerous close substitutes in the canned tuna market.

f. Test the hypothesis that elasticity of sales of brand no. 1 with respect to the price of brand no. 1 from part (e) is minus three against the alternative that the elasticity is not minus three. Use the 10% level of significance. Clearly, state the null and alternative hypotheses in terms of the model parameters, give the rejection region, and the p-value for the test. What is your conclusion?

H_0: elasticity $= -3$, H_1: elasticity $\neq -3$

$t_0 = -2.25$, $\alpha = 0.1$

reject region = $t > t_{(0.05, 48)} = 1.6759$

p-value $= 2P(t > |t_0|) = 0.0288 < 0.1 \Rightarrow$ reject H_0.

Decision: Since the p–value is less than the significance level (alpha=0.10), we reject the null hypothesis.
Conclusion: There is sufficient evidence to conclude that the price elasticity of demand for this brand of tuna is significantly different from –3. The data suggests that consumers are even more price–sensitive than hypothesized.